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Thermophysical properties of propan-2-ol (isopropyl alcohol):

Part 1. The ideal gas and the saturated liquid and vapour

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THE PREPARATION OF THIS DATA ITEM

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THERMOPHYSICAL PROPERTIES OF PROPAN-2-OL (ISOPROPYL ALCOHOL): PART 1. THE IDEAL GAS AND THE SATURATED LIQUID AND VAPOUR

CONTENTS

		Page
1.	INTRODUCTION	1
2.	NOTATION AND UNITS	1
3.	THERMOPHYSICAL PROPERTIES	2
4.	EQUATIONS FOR PROPERTIES	3
	4.1 Vapour Pressure	3
	4.2 Density of the Saturated Liquid	3
	4.3 Density of the Saturated Vapour	3
	4.4 Specific Enthalpy of the Saturated Liquid	3
	4.5 Latent Heat of Vaporisation (Specific Enthalpy of Vaporisation)	3
	4.6 Specific Enthalpy of the Saturated Vapour	3
	4.7 Isobaric Specific Heat Capacity of the Saturated Liquid	3
	4.8 Isobaric Specific Heat Capacity of the Saturated Vapour	4
	4.9 Dynamic Viscosity of the Saturated Liquid	4
	4.10 Dynamic Viscosity of the Saturated Vapour	4
	4.11 Thermal Conductivity of the Saturated Liquid	4
	4.12 Thermal Conductivity of the Saturated Vapour	4
	4.13 Surface Tension	4
	4.14 Isobaric Specific Heat Capacity of the Ideal Gas	4
	4.15 Specific Enthalpy of the Ideal Gas	5
	4.16 Dynamic Viscosity of the Dilute Gas	5
	4.17 Thermal Conductivity of the Dilute Gas	5
5.	EXPLANATION OF THE TABLES	5
6.	REFERENCES AND DERIVATION	6
7.	TABLES	9
	TABLE 7.1 Physical constants of propan-2-ol and constants for Equations (4.1) to (4.17)	9
	TABLE 7.2 Thermophysical properties of propan-2-ol on the saturation line for temperature in degree Celsius	10
	TABLE 7.3 Thermophysical properties of propan-2-ol on the saturation line for temperature in kelvin	12

TABLE 7.4 Thermophysical properties of propan-2-ol in the ideal and dilute gas for temperature in degree Celsius 14

TABLE 7.5 Thermophysical properties of propan-2-ol in the ideal and dilute gas for temperature in kelvin 15

APPENDIX A CONVERSION FACTORS 16

APPENDIX B 17

B1. SPECIFIC HEAT CAPACITIES OF SATURATED LIQUIDS AND VAPOURS 17

THERMOPHYSICAL PROPERTIES OF PROPAN-2-OL (ISOPROPYL ALCOHOL): PART 1. THE IDEAL GAS AND THE SATURATED LIQUID AND VAPOUR

1. INTRODUCTION

This Item, which is one of a series concerned with the thermophysical properties of industrially important fluids, gives equations representing the thermophysical properties of propan-2-ol on the saturation line, and in the ideal gas state and dilute gas state, together with tabulated values calculated from the equations.

The data on which the recommended values of the fluid properties are based are considered to be the best available and the range covered for the saturation line properties is from the melting temperature to within at least 56 kelvin of the critical temperature. The properties of the ideal and dilute gas are given for temperatures from 185 to 1000 K. Properties in the single phase gas and liquid states are given in Part 2.

All quantities are expressed in SI units or their multiples (see Section 2) but, for users who require them in other units, conversion factors are provided in Appendix A. Equations representing the fluid properties are given in Section 4 and Section 5 gives an explanation of the Tables in the Data Item.

2. NOTATION AND UNITS

b_1, b_2, b_3, b_4, b_5	constants in Equations (4.1) to (4.17)	*
c	specific heat capacity	kJ/kg K
h	specific enthalpy	kJ/kg
M	molecular weight (relative molecular mass)	–
p	vapour pressure	kPa
T	kelvin temperature	K
t	Celsius temperature, (T – 273.15)	ºC
u	specific internal energy	kJ/kg
V	molar volume	cm^3/mol
v	specific volume, $v = 1/\rho$	m^3/kg
η	dynamic viscosity	mN s/m^2 or µN s/m^2
θ	temperature dependent variable in Equation (4.9)	–
λ	thermal conductivity	mW/m K
ρ	density	kg/m^3
σ	surface tension	mN/m
ω	acentric factor	–

* The constants b are dimensionless except for the following: Equations (4.4) and (4.6), kJ/kg (all); Equations (4.7) and (4.8), kJ/kg K (b_1); Equation (4.9), K (b_3), mN s/m^2 (b_4); Equation (4.10), µN s/m^2 (exp(b_1)); Equations (4.11) and (4.12), mW/m K ($b_1$and exp(b_1) respectively); Equation (4.13), mN/m (b_1); Equation (4.14), kJ/kg K (all); Equation (4.15), kJ/kg (all); Equation (4.16), µN s/m^2 (all); Equation (4.17), mW/m K (all).

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State descriptors

(g)	value for gas or saturated vapour
(l)	value for saturated liquid
Δ	denotes property difference between two states

Superscripts

id	denotes ideal gas
(o)	denotes dilute gas

Subscripts

b	value at normal boiling point
c	value at critical point
m	value at normal melting point
p	value at constant pressure
r	denotes reduced quantity, *e.g.* $T_r = T/T_c$
sat	value at saturation (defined in Appendix B, Section B1)
v	value at constant volume
vap	denotes evaporation.

3. THERMOPHYSICAL PROPERTIES

Equations for the ideal gas specific heat capacity and specific enthalpy were fitted to values derived from those published by the Thermodynamics Research Center (Derivation 31).

Tables for the thermodynamic properties at saturation and selected pressure intervals were issued in an earlier Data Item (Derivation 21)*. These were extended down to the melting temperature using a computer software package, codename 'LOADER', developed by the National Engineering Laboratory (Derivation 30) to maintain a thermodynamically consistent set of results. The liquid heat capacity at constant pressure was evaluated by the relationship defined in Appendix B, Section B1.

The equation for the viscosity of the vapour in the dilute gas was fitted to values derived from Derivation 17 which were supported by Derivations 10 and 20. The equation for the thermal conductivity of the vapour in the dilute gas was fitted to values based on data from Derivation 24 which were supported by Derivations 11 and 16. Values for these properties in the saturated vapour were determined from the dilute gas data by the software package, 'LOADER' and Equations (4.10) and (4.12) were fitted to the results.

The surface tension data (Reference 2) were extended by the package (Derivation 30) .

Equations for the liquid thermal conductivity, vapour pressure and liquid viscosity were reported in previous Data Items (Derivations 27, 29, and 28). Equations (4.1) to (4.17) derived from the above sources were used to generate the property values in Tables 7.2 to 7.5.

* Reissued in Table 6.1 of Data Item 89020.

4. EQUATIONS FOR PROPERTIES

4.1 Vapour Pressure

Data for the vapour pressure are represented by Equation (4.1).

$$\log_e\left(\frac{p}{p_c}\right) = \frac{1}{T_r}[b_1(1-T_r) + b_2(1-T_r)^{3/2} + b_3(1-T_r)^{5/2} + b_4(1-T_r)^5]\,. \tag{4.1}$$

4.2 Density of the Saturated Liquid

Data for the density of the saturated liquid are represented by Equation (4.2).

$$\log_e\left(\frac{\rho(\mathrm{l})}{\rho_c}\right) = b_1(1-T_r)^{1/3} + b_2(1-T_r)^{2/3} + b_3(1-T_r) + b_4(1-T_r)^{4/3}\,. \tag{4.2}$$

4.3 Density of the Saturated Vapour

Data for the density of the saturated vapour are represented by Equation (4.3).

$$\log_e\left(\frac{\rho(\mathrm{g})}{\rho_c}\right) = b_1\left(\frac{1}{T_r}-1\right)^{1/3} + b_2\left(\frac{1}{T_r}-1\right)^{2/3} + b_3\left(\frac{1}{T_r}-1\right) + b_4\left(\frac{1}{T_r}-1\right)^{4/3} + b_5\left(\frac{1}{T_r}-1\right)^{5/3}. \tag{4.3}$$

4.4 Specific Enthalpy of the Saturated Liquid

Data for the specific enthalpy of the saturated liquid are represented by Equation (4.4).

$$h(\mathrm{l}) = b_1 + b_2(1-T_r)^{1/3} + b_3(1-T_r)^{2/3} + b_4(1-T_r) + b_5(1-T_r)^{4/3}\,. \tag{4.4}$$

4.5 Latent Heat of Vaporisation (Specific Enthalpy of Vaporisation)

Data for the latent heat of vaporisation are represented by Equation (4.5).

$$\Delta_{\mathrm{vap}}h = h(\mathrm{g}) - h(\mathrm{l})\,. \tag{4.5}$$

4.6 Specific Enthalpy of the Saturated Vapour

Data for the specific enthalpy of the saturated vapour are represented by Equation (4.6).

$$h(\mathrm{g}) = b_1 + b_2(1-T_r)^{1/3} + b_3(1-T_r)^{2/3} + b_4(1-T_r) + b_5(1-T_r)^{4/3}\,. \tag{4.6}$$

4.7 Isobaric Specific Heat Capacity of the Saturated Liquid

Data for the isobaric specific heat capacity of the saturated liquid are represented by Equation (4.7).

$$c_p(\mathrm{l}) = b_1[1 + b_2(1-T_r)^{-1} + b_3(1-T_r) + b_4(1-T_r)^2 + b_5(1-T_r)^3]\,. \tag{4.7}$$

The significance of $c_p(\mathrm{l})$ as a representation of specific heat capacity is fully explained in Appendix B, Section B1.

4.8 Isobaric Specific Heat Capacity of the Saturated Vapour

Data for the isobaric specific heat capacity of the saturated vapour are represented by Equation (4.8).

$$c_p(\mathrm{g}) = b_1[1 + b_2(1-T_\mathrm{r})^{-2/3} + b_3(1-T_\mathrm{r})^{-1/3} + b_4(1-T_\mathrm{r})^{1/3} + b_5(1-T_\mathrm{r})^{2/3}]\,. \tag{4.8}$$

The significance of $c_p(\mathrm{g})$ as a representation of specific heat capacity is fully explained in Appendix B, Section B1.

4.9 Dynamic Viscosity of the Saturated Liquid

Data for the dynamic viscosity of the saturated liquid are represented by Equation (4.9).

$$\eta(\mathrm{l}) = b_4 \exp[b_1(\theta-1)^{1/3} + b_2(\theta-1)^{4/3}] \tag{4.9}$$

where $\theta = (T_\mathrm{c} - b_3)/(T - b_3)\,.$

4.10 Dynamic Viscosity of the Saturated Vapour

Data for the dynamic viscosity of the saturated vapour are represented by Equation (4.10).

$$\log_\mathrm{e}\,\eta(\mathrm{g}) = b_1 + b_2\left(\frac{1}{T_\mathrm{r}} - 1\right)^{1/3} + b_3\left(\frac{1}{T_\mathrm{r}} - 1\right)^{4/3}. \tag{4.10}$$

4.11 Thermal Conductivity of the Saturated Liquid

Data for the thermal conductivity of the saturated liquid are represented by Equation (4.11).

$$\lambda(\mathrm{l}) = b_1[1 + b_2(1-T_\mathrm{r})^{1/3} + b_3(1-T_\mathrm{r})^{2/3} + b_4(1-T_\mathrm{r})]\,. \tag{4.11}$$

4.12 Thermal Conductivity of the Saturated Vapour

Data for the thermal conductivity of the saturated vapour are represented by Equation (4.12).

$$\log_\mathrm{e}\,\lambda(\mathrm{g}) = b_1 + b_2(1-T_\mathrm{r})^{-2/3} + b_3(1-T_\mathrm{r})^{-1/3} + b_4(1-T_\mathrm{r})^{1/3} + b_5(1-T_\mathrm{r})^{2/3}\,. \tag{4.12}$$

4.13 Surface Tension

Data for surface tension are represented by Equation (4.13).

$$\sigma = b_1(1-T_\mathrm{r})^{b_2}[1 + b_3(1-T_\mathrm{r})]\,. \tag{4.13}$$

4.14 Isobaric Specific Heat Capacity of the Ideal Gas

Data for the isobaric specific heat capacity of the ideal gas are represented by Equation (4.14).

$$c_p^{\mathrm{id}}(\mathrm{g}) = b_1 + b_2 T_\mathrm{r}^{-1/2} + b_3 T_\mathrm{r}^{1/2} + b_4 T_\mathrm{r}\,. \tag{4.14}$$

4.15 Specific Enthalpy of the Ideal Gas

Data for the specific enthalpy of the ideal gas are represented by Equation (4.15).

$$h^{\mathrm{id}}(\mathrm{g}) = b_1 + b_2 T_\mathrm{r}^{1/2} + b_3 T_\mathrm{r} + b_4 T_\mathrm{r}^{3/2} + b_5 T_\mathrm{r}^2 . \tag{4.15}$$

(Equations (4.14) and (4.15) are thermodynamically consistent.)

4.16 Dynamic Viscosity of the Dilute Gas

Data for the dynamic viscosity of the dilute gas are represented by Equation (4.16).

$$\eta^{(\mathrm{o})}(\mathrm{g}) = b_1 + b_2 T_\mathrm{r} + b_3 T_\mathrm{r}^2 + b_4 T_\mathrm{r}^3 . \tag{4.16}$$

4.17 Thermal Conductivity of the Dilute Gas

Data for the thermal conductivity of the dilute gas are represented by Equation (4.17).

$$\lambda^{(\mathrm{o})}(\mathrm{g}) = b_1 + b_2 T_\mathrm{r} + b_3 T_\mathrm{r}^2 + b_4 T_\mathrm{r}^3 + b_5 T_\mathrm{r}^4 . \tag{4.17}$$

5. EXPLANATION OF THE TABLES

Table 7.1 provides values of the physical constants and constants for use in the equations given in Section 4, that represent the fluid properties in the SI units* of the Notation, Section 2. Modern theory requires the isobaric heat capacity, viscosity and thermal conductivity to tend to infinity as the critical point is approached; this theoretical limit has been included where appropriate in Tables 7.2 and 7.3. However, Equations (4.9), (4.10), (4.11) and (4.12) do not conform with theory in this respect; Equations (4.9), (4.10) and (4.11) approach a finite value and Equation (4.12) goes to zero at the critical temperature. Although the remaining equations, Equations (4.1), (4.2), (4.3), (4.4), (4.5), (4.6), (4.7), (4.8) and (4.13), applying on the saturation line extrapolate in a physically realistic manner, users should not extrapolate any of the equations in Section 4 outside the temperature limits specified in Table 7.1.

The values derived from the equations applying on the saturation line, Tables 7.2 and 7.3, are presented from the melting temperature to approximately $T_\mathrm{r} = 0.98$, except for Equation (4.11) which terminates at $T_\mathrm{r} = 0.89$ and Equations (4.1), (4.2), (4.3), (4.4), (4.5), (4.6) and (4.13) which are valid to the critical temperature. The specific enthalpy values are calculated on the basis that the enthalpy of the crystalline solid is zero at 0 K.

The specific enthalpies of vaporisation in column 7 of Tables 7.2 and 7.3 were calculated directly from Equation (4.5) with subsequent rounding to the desired number of significant figures. Small differences, usually in the last figure may occur therefore, between these values and those obtained by subtraction of the specific ethalpies in column 5 from those in column 6, since the values generated thereby are the differences between two rounded numbers.

Tables 7.4 and 7.5 present values for the ideal and dilute gas from 185.2 to 1000 K.

Parentheses are used in all the tables to indicate the values not directly supported by experimental data.

* Appendix A gives conversion factors to other units.

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The references given are recommended sources of information supplementary to that in this Data Item.

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7. TABLES

C_3H_8O PROPAN-2-OL (ISOPROPYL ALCOHOL)
CAS Reg. No: 67-63-0

TABLE 7.1 Physical constants of propan-2-ol and constants for Equations (4.1) to (4.17)

Physical Constants	*Symbol*	*Value*	
Molecular Weight	M	60.1	
Critical Temperature	T_c	508.3 K	(235.15 ^oC)
Critical Pressure	p_c	4763. kPa	
Critical Density	ρ_c	273.2 kg/m^3	
Critical Molar Volume	V_c	220. cm^3/mol	
Normal Boiling Temperature	T_b	355.39 K	(82.24 ^oC)
Normal Melting Temperature	T_m	185.2 K	(-87.95 ^oC)
Acentric Factor	ω	0.665	

Constants used in Equations (4.1) to (4.13)

Property	Equation in Section 4	b_1	b_2	b_3	b_4	b_5	Range of Applicability ^oC
p	(4.1)	-8.74110	2.16825	-8.6964	4.680	-	-88 to 235
ρ(l)	(4.2)	1.80821	-0.69176	0.63646	-0.57154	-	-88 to 235
ρ(g)	(4.3)	-0.367389	-13.5875	26.5406	-31.7279	8.43213	-88 to 235
h(l)	(4.4)	1342.60	-814.972	3766.54	-10014.82	5814.38	-88 to 235
h(g)	(4.6)	1342.60	399.098	117.736	-1570.63	559.689	-88 to 235
c_p(l)	(4.7)	3.78653	0.007907	1.30013	-8.30586	7.79943	-88 to 225
c_p(g)	(4.8)	23.7390	0.081642	-0.417372	-0.901198	0.265128	-88 to 225
η(l)	(4.9)	2.546	2.527	45.0	0.02573	-	-88 to 220
η(g)	(4.10)	3.0587	-1.03174	-0.108725	-	-	-88 to 225
λ(l)	(4.11)	72.96	3.0	-6.5	5.4	-	-88 to 180
λ(g)	(4.12)	-74.2483	-3.40739	26.8325	100.263	-49.8073	-88 to 202
σ	(4.13)	41.534	0.7749	0	-	-	-88 to 235

Constants used in Equations (4.14) to (4.17)

c_p^{id}(g)	(4.14)	-8.34651	2.18702	11.5437	-3.17024	-	-88 to 727
h^{id} (g)	(4.15)	534.037	2223.32	-4242.53	3911.78	-805.716	-88 to 727
$\eta^{(o)}$(g)	(4.16)	-0.257051	14.4652	-0.675850	-0.203697	-	-88 to 727
$\lambda^{(o)}$(g)	(4.17)	-0.133292	-4.37009	62.8279	-23.5512	3.36545	-88 to 727

TABLE 7.2
Thermophysical properties of propan-2-ol on the saturation line for temperature in degree Celsius

t	p	ρ(l)	ρ(g)	h(l)	h(g)	$\Delta_{vap}h$
°C	kPa	kg/m^3	kg/m^3	kJ/kg	kJ/kg	kJ/kg
-87.95	0.0000	(850.4)	(0.00000)	(238.)	(1080.)	(842.)
-80	0.0001	(846.9)	(0.00000)	251.	(1091.)	(840.)
-70	0.0007	(842.1)	(0.00002)	268.	(1104.)	(836.)
-60	0.0026	(837.1)	(0.00008)	286.	(1117.)	(831.)
-50	0.0091	(831.8)	(0.00028)	304.	(1130.)	(825.)
-40	0.0279	(826.2)	(0.00084)	324.	(1143.)	(819.)
-30	0.0780	820.2	(0.00226)	345.	(1156.)	(811.)
-20	0.1997	813.9	(0.00559)	366.	(1169.)	(803.)
-10	0.4726	807.2	0.01279	389.	(1182.)	(793.)
0	1.043	800.2	0.02728	413.	(1195.)	783.
10	2.160	792.8	0.05467	437.	(1209.)	771.
20	4.226	785.0	0.1036	463.	(1222.)	759.
30	7.850	776.7	0.1869	490.	(1235.)	745.
40	13.91	768.1	0.3224	517.	(1248.)	731.
50	23.62	758.9	0.5340	546.	(1261.)	715.
60	38.56	749.3	0.8531	576.	(1274.)	698.
70	60.74	739.1	1.319	607.	(1287.)	680.
80	92.59	728.4	1.980	639.	(1300.)	661.
82.24	101.325	725.9	2.160	647.	(1303.)	656.
90	137.0	717.1	2.893	673.	(1313.)	640.
100	197.2	705.2	4.127	707.	(1326.)	619.
105	234.4	698.9	4.888	725.	(1332.)	607.
110	276.9	692.5	5.761	743.	(1339.)	596.
115	325.2	685.9	6.755	761.	(1345.)	584.
120	379.8	679.0	7.885	780.	(1351.)	571.
125	441.3	672.0	9.165	799.	(1357.)	559.
130	510.2	664.7	10.61	818.	(1363.)	546.
135	587.0	657.2	12.23	837.	(1369.)	532.
140	672.3	649.5	14.05	857.	(1375.)	518.
145	766.6	641.4	16.10	877.	(1381.)	504.
150	870.4	633.1	18.38	897.	(1387.)	490.
155	984.4	624.4	20.92	918.	(1393.)	475.
160	1109.	615.4	23.77	939.	(1398.)	459.
165	1245.	606.1	26.93	960.	(1403.)	444.
170	1393.	596.3	30.47	981.	(1409.)	427.
175	1553.	586.0	34.43	1003.	(1414.)	411.
180	1727.	575.2	38.85	1025.	(1418.)	393.
185	1914.	563.8	43.83	1047.	(1423.)	375.
190	2116.	551.8	49.45	1070.	(1427.)	357.
195	2332.	538.9	55.85	1092.	(1430.)	338.
200	2565.	525.1	63.19	1115.	(1434.)	318.
205	2815.	510.2	71.73	1138.	(1436.)	298.
210	3083.	493.7	81.85	(1161.)	(1437.)	276.
215	3370.	475.2	94.15	(1184.)	(1438.)	253.
220	3678.	454.0	109.7	(1207.)	(1436.)	229.
225	4009.	428.1	130.7	(1230.)	(1431.)	201.
230	4365.	393.2	163.2	(1254.)	(1420.)	166.
235	4751.	307.3	252.8	(1302.)	(1369.)	67.0
235.15	4763.	273.2	273.2	(1343.)	(1343.)	0

TABLE 7.2 (concluded)
Thermophysical properties of propan-2-ol on the saturation line for temperature in degree Celsius

t °C	c_p(l) kJ/kg K	c_p(g) kJ/kg K	η(l) mN s/m^2	η(g) μN s/m^2	λ(l) mW/m K	λ(g) mW/m K	σ mN/m
-87.95	(1.840)	(1.096)	(1628.1)	(4.896)	(161.0)	(5.674)	(29.24)
-80	1.836	(1.119)	(683.83)	(5.131)	159.1	(6.126)	(28.68)
-70	1.847	(1.149)	(264.98)	(5.420)	156.6	(6.733)	(27.97)
-60	1.875	(1.180)	(116.94)	(5.703)	154.3	(7.383)	(27.26)
-50	1.918	(1.212)	(57.368)	5.980	151.9	(8.076)	(26.54)
-40	1.975	(1.245)	(30.704)	6.251	149.6	(8.814)	(25.81)
-30	2.046	(1.279)	17.667	6.519	147.4	(9.596)	(25.08)
-20	2.128	(1.314)	10.803	6.783	145.1	(10.42)	(24.35)
-10	2.220	(1.350)	6.9537	7.044	143.0	(11.29)	(23.60)
0	2.322	(1.387)	4.6766	7.302	140.8	(12.19)	22.86
10	2.431	(1.426)	3.2657	7.559	138.8	(13.13)	22.10
20	2.546	(1.466)	2.3556	7.815	136.7	(14.10)	21.33
30	2.666	(1.508)	1.7476	8.071	134.7	(15.11)	20.56
40	2.790	(1.551)	1.3287	8.328	132.8	(16.13)	19.78
50	2.916	(1.597)	1.0321	8.585	130.9	(17.18)	18.99
60	3.044	(1.645)	0.8169	8.845	129.1	(18.24)	18.19
70	3.171	(1.695)	0.6573	9.108	127.3	(19.32)	17.38
80	3.297	(1.748)	0.5366	9.375	125.6	(20.40)	16.56
82.24	3.325	(1.760)	0.5137	9.436	125.2	(20.64)	(16.37)
90	3.420	(1.804)	0.4437	9.647	124.0	(21.48)	(15.73)
100	3.540	(1.864)	0.3710	9.926	122.4	(22.57)	(14.88)
105	3.598	(1.895)	0.3405	10.07	121.6	(23.12)	(14.45)
110	3.654	(1.928)	0.3133	10.21	120.9	(23.67)	(14.02)
115	3.710	(1.963)	0.2888	10.36	120.1	(24.23)	(13.58)
120	3.763	(1.998)	0.2668	10.51	119.4	(24.79)	(13.14)
125	3.815	(2.036)	0.2469	10.66	118.7	(25.36)	(12.70)
130	3.865	(2.076)	0.2288	10.82	118.0	(25.94)	(12.25)
135	3.913	(2.118)	0.2124	10.98	117.4	(26.54)	(11.80)
140	3.960	(2.163)	0.1974	11.14	116.7	(27.15)	(11.34)
145	4.004	(2.211)	0.1838	11.31	116.1	(27.80)	(10.87)
150	4.046	(2.263)	0.1712	11.49	115.5	(28.48)	(10.40)
155	4.086	(2.319)	0.1596	11.67	114.9	(29.20)	(9.926)
160	4.125	(2.380)	0.1489	11.85	114.3	(29.97)	(9.443)
165	4.161	(2.449)	0.1391	12.05	113.8	(30.80)	(8.952)
170	4.197	(2.525)	0.1299	12.25	113.3	(31.72)	(8.453)
175	4.231	(2.612)	0.1213	12.47	112.7	(32.73)	(7.946)
180	4.264	(2.713)	0.1132	12.69	112.2	(33.84)	(7.429)
185	4.298	(2.832)	0.1057	12.93		(35.06)	(6.902)
190	4.333	(2.976)	0.0985	13.18		(36.38)	(6.362)
195	4.373	(3.154)	0.0917	13.46		(37.72)	(5.809)
200	4.419	(3.383)	0.0851	13.76		(38.92)	(5.240)
205	4.479	(3.690)	0.0788	14.09			(4.653)
210	(4.562)	(4.125)	0.0727	14.46			(4.043)
215	(4.689)	(4.788)	0.0665	14.89			(3.405)
220	(4.911)	(5.916)	0.0603	15.40			(2.730)
225	(5.372)	(8.209)		16.06			(2.001)
230							(1.183)
235							(0.076)
235.15	∞	∞	∞	∞	∞	∞	0

TABLE 7.3
Thermophysical properties of propan-2-ol on the saturation line
for temperature in kelvin

T	p	ρ(l)	ρ(g)	h(l)	h(g)	$\Delta_{vap}h$
K	kPa	kg/m^3	kg/m^3	kJ/kg	kJ/kg	kJ/kg
185.2	0.0000	(850.4)	(0.00000)	(238.)	(1080.)	(842.)
190	0.0001	(848.3)	(0.00000)	(246.)	(1087.)	(841.)
200	0.0004	(843.7)	(0.00001)	262.	(1100.)	(837.)
210	0.0017	(838.7)	(0.00006)	280.	(1113.)	(833.)
220	0.0062	(833.5)	(0.00019)	298.	(1126.)	(827.)
230	0.0198	(828.0)	(0.00060)	318.	(1139.)	(821.)
240	0.0570	(822.1)	(0.00167)	338.	(1152.)	(814.)
250	0.1499	815.9	(0.00424)	359.	(1165.)	(805.)
260	0.3632	809.4	(0.00994)	382.	(1178.)	(796.)
270	0.8186	802.4	0.02164	405.	(1191.)	(786.)
280	1.729	795.2	0.04420	429.	(1204.)	775.
290	3.441	787.5	0.08520	455.	(1218.)	763.
300	6.493	779.4	0.1560	481.	(1231.)	750.
310	11.67	770.8	0.2727	509.	(1244.)	735.
320	20.08	761.9	0.4574	537.	(1257.)	720.
330	33.17	752.4	0.7388	567.	(1270.)	703.
340	52.83	742.4	1.154	597.	(1283.)	686.
350	81.35	731.9	1.747	629.	(1296.)	667.
355	99.77	726.4	2.128	646.	(1303.)	657.
355.39	101.325	725.9	2.160	647.	(1303.)	656.
360	121.5	720.7	2.574	662.	(1309.)	647.
365	146.8	715.0	3.095	679.	(1316.)	636.
370	176.3	709.0	3.699	696.	(1322.)	626.
375	210.4	702.9	4.397	714.	(1328.)	615.
380	249.4	696.6	5.198	732.	(1335.)	603.
385	294.0	690.1	6.114	750.	(1341.)	591.
390	344.6	683.4	7.157	768.	(1347.)	579.
395	401.7	676.5	8.341	787.	(1353.)	567.
400	465.9	669.3	9.679	806.	(1360.)	554.
405	537.7	662.0	11.19	825.	(1366.)	541.
410	617.6	654.4	12.88	844.	(1372.)	527.
415	706.1	646.5	14.78	864.	(1378.)	513.
420	803.9	638.4	16.91	884.	(1383.)	499.
425	911.4	629.9	19.29	905.	(1389.)	484.
430	1029.	621.2	21.94	926.	(1395.)	469.
435	1158.	612.0	24.90	946.	(1400.)	454.
440	1298.	602.5	28.20	968.	(1405.)	438.
445	1451.	592.5	31.88	989.	(1410.)	421.
450	1616.	582.1	36.00	1011.	(1415.)	404.
455	1794.	571.1	40.63	1033.	(1420.)	387.
460	1987.	559.5	45.83	1056.	(1424.)	369.
465	2194.	547.1	51.72	1078.	(1428.)	350.
470	2417.	533.9	58.44	1101.	(1432.)	331.
475	2656.	519.7	66.19	1124.	(1435.)	311.
480	2912.	504.3	75.26	1147.	(1437.)	290.
485	3187.	487.1	86.10	(1170.)	(1438.)	268.
490	3482.	467.8	99.44	(1193.)	(1437.)	245.
495	3798.	445.1	116.7	(1216.)	(1435.)	219.
500	4137.	416.7	140.8	(1239.)	(1428.)	189.
505	4504.	375.0	181.9	(1264.)	(1412.)	148.
508.3	4763.	273.2	273.2	(1343.)	(1343.)	0

TABLE 7.3 (concluded)
Thermophysical properties of propan-2-ol on the saturation line for temperature in kelvin

T K	c_p(l) kJ/kg K	c_p(g) kJ/kg K	η(l) mN s/m^2	η(g) μN s/m^2	λ(l) mW/m K	λ(g) mW/m K	σ mN/m
185.2	(1.840)	(1.096)	(1628.1)	(4.896)	(161.0)	(5.674)	(29.24)
190	(1.836)	(1.110)	(951.60)	(5.038)	(159.8)	(5.944)	(28.90)
200	1.842	(1.139)	(351.75)	(5.330)	157.4	(6.537)	(28.19)
210	1.864	(1.170)	(149.44)	(5.614)	155.0	(7.173)	(27.48)
220	1.903	(1.202)	(71.068)	(5.893)	152.7	(7.853)	(26.76)
230	1.956	(1.234)	(37.072)	6.166	150.3	(8.577)	(26.04)
240	2.022	(1.268)	(20.879)	6.435	148.1	(9.345)	(25.31)
250	2.101	(1.302)	12.538	6.700	145.8	(10.16)	(24.58)
260	2.190	(1.338)	7.9485	6.962	143.6	(11.01)	(23.84)
270	2.289	(1.375)	5.2762	7.221	141.5	(11.90)	(23.09)
280	2.396	(1.413)	3.6432	7.478	139.4	(12.83)	22.34
290	2.509	(1.453)	2.6025	7.735	137.4	(13.79)	21.58
300	2.628	(1.495)	1.9146	7.991	135.3	(14.79)	20.81
310	2.751	(1.538)	1.4450	8.247	133.4	(15.81)	20.03
320	2.876	(1.582)	1.1152	8.504	131.5	(16.85)	19.24
330	3.004	(1.629)	0.8777	8.763	129.6	(17.91)	18.44
340	3.131	(1.679)	0.7027	9.025	127.9	(18.98)	17.64
350	3.258	(1.731)	0.5712	9.291	126.1	(20.06)	16.82
355	3.320	(1.758)	0.5176	9.425	125.3	(20.60)	(16.41)
355.39	3.325	(1.760)	0.5137	9.436	125.2	(20.64)	(16.37)
360	3.382	(1.786)	0.4705	9.561	124.5	(21.14)	(15.99)
365	3.443	(1.814)	0.4289	9.699	123.7	(21.68)	(15.57)
370	3.503	(1.844)	0.3921	9.838	122.9	(22.23)	(15.15)
375	3.561	(1.875)	0.3593	9.979	122.1	(22.77)	(14.72)
380	3.619	(1.907)	0.3301	10.12	121.3	(23.32)	(14.29)
385	3.675	(1.941)	0.3039	10.27	120.6	(23.88)	(13.86)
390	3.730	(1.976)	0.2804	10.42	119.9	(24.43)	(13.42)
395	3.783	(2.012)	0.2592	10.57	119.1	(25.00)	(12.98)
400	3.834	(2.051)	0.2400	10.72	118.5	(25.57)	(12.53)
405	3.883	(2.091)	0.2226	10.88	117.8	(26.16)	(12.08)
410	3.931	(2.134)	0.2067	11.04	117.1	(26.76)	(11.63)
415	3.976	(2.180)	0.1922	11.21	116.5	(27.39)	(11.17)
420	4.020	(2.230)	0.1790	11.38	115.9	(28.04)	(10.70)
425	4.061	(2.283)	0.1668	11.55	115.3	(28.74)	(10.23)
430	4.101	(2.341)	0.1556	11.74	114.7	(29.47)	(9.748)
435	4.139	(2.405)	0.1452	11.93	114.1	(30.27)	(9.262)
440	4.175	(2.476)	0.1356	12.12	113.6	(31.13)	(8.768)
445	4.209	(2.556)	0.1266	12.33	113.1	(32.08)	(8.267)
450	4.243	(2.648)	0.1182	12.55	112.5	(33.13)	(7.756)
455	4.277	(2.755)	0.1104	12.78		(34.28)	(7.236)
460	4.311	(2.882)	0.1030	13.02		(35.54)	(6.704)
465	4.347	(3.037)	0.0959	13.28		(36.87)	(6.160)
470	4.389	(3.232)	0.0892	13.57		(38.19)	(5.601)
475	4.439	(3.486)	0.0828	13.88		(39.25)	(5.025)
480	4.506	(3.832)	0.0765	14.22			(4.430)
485	(4.602)	(4.336)	0.0704	14.61			(3.811)
490	(4.756)	(5.131)	0.0642	15.07			(3.160)
495	(5.039)	(6.558)		15.62			(2.468)
500							(1.712)
505							(0.838)
508.3	∞	∞	∞	∞	∞	∞	0

TABLE 7.4
Thermophysical properties of propan-2-ol in the ideal and dilute gas for temperature in degree Celsius

	Ideal gas		Dilute gas	
t	c_p^{id}(g)	h^{id}(g)	$\eta^{(o)}$(g)	$\lambda^{(o)}$(g)
°C	kJ/kg K	kJ/kg	μN s/m^2	mW/m K
-87.95	(1.090)	(1084.)	(4.914)	(5.535)
-80	1.113	1092.	(5.131)	(6.056)
-60	1.177	1115.	(5.675)	(7.450)
-40	1.247	1140.	(6.216)	(8.957)
-20	1.320	1165.	(6.754)	(10.57)
0	1.396	1192.	7.289	(12.29)
20	1.472	1221.	7.822	(14.10)
25	1.491	1228.	7.954	(14.57)
40	1.547	1251.	8.350	16.00
60	1.623	1283.	8.876	17.98
80	1.697	1316.	9.398	20.04
100	1.769	1351.	9.917	22.18
120	1.840	1387.	10.43	24.38
140	1.910	1424.	10.94	26.64
160	1.977	1463.	11.45	28.97
180	2.043	1503.	11.96	31.34
200	2.107	1545.	12.46	33.77
220	2.168	1588.	12.95	36.24
240	2.228	1632.	13.45	38.75
260	2.286	1677.	13.94	41.30
280	2.342	1723.	14.42	43.88
300	2.396	1770.	14.90	46.50
320	2.449	1819.	15.38	(49.14)
340	2.499	1868.	15.85	(51.80)
360	2.548	1919.	16.32	(54.49)
380	2.595	1970.	16.78	(57.20)
400	2.640	2023.	17.24	(59.92)
420	2.683	2076.	17.70	(62.66)
440	2.725	2130.	18.14	(65.41)
460	2.766	2185.	18.59	(68.17)
480	2.804	2241.	19.03	(70.94)
500	2.842	2297.	19.46	(73.71)
520	2.877	2354.	(19.89)	(76.50)
540	2.912	2412.	(20.32)	(79.29)
560	2.944	2471.	(20.74)	(82.08)
580	2.976	2530.	(21.15)	(84.88)
600	3.006	2590.	(21.56)	(87.68)
620	3.035	2650.	(21.97)	(90.48)
640	3.062	2711.	(22.37)	(93.29)
660	3.088	2773.	(22.76)	(96.10)
680	3.113	2835.	(23.15)	(98.92)
700	3.137	2897.	(23.53)	(101.7)
720	3.160	2960.	(23.91)	(104.6)

TABLE 7.5
Thermophysical properties of propan-2-ol in the ideal and dilute gas for temperature in kelvin

	Ideal gas		Dilute gas	
T	c_p^{id}(g)	h^{id}(g)	$\eta^{(o)}$(g)	$\lambda^{(o)}$(g)
K	kJ/kg K	kJ/kg	μN s/m^2	mW/m K
185.2	(1.090)	(1084.)	(4.914)	(5.535)
200	1.134	1100.	(5.318)	(6.520)
220	1.200	1123.	(5.861)	(7.953)
240	1.272	1148.	(6.401)	(9.498)
260	1.346	1174.	(6.938)	(11.15)
280	1.422	1202.	7.472	(12.90)
298.15	1.491	1228.	7.954	(14.57)
300	1.498	1231.	8.003	14.74
320	1.573	1262.	8.531	16.67
340	1.648	1294.	9.055	18.68
360	1.722	1328.	9.576	20.77
380	1.794	1363.	10.09	22.92
400	1.864	1400.	10.61	25.15
420	1.933	1438.	11.12	27.43
440	2.000	1477.	11.63	29.78
460	2.065	1518.	12.13	32.17
480	2.128	1559.	12.63	34.61
500	2.189	1603.	13.12	37.10
520	2.248	1647.	13.62	39.62
540	2.306	1693.	14.10	42.18
560	2.361	1739.	14.59	44.78
580	2.414	1787.	15.07	(47.40)
600	2.466	1836.	15.54	(50.05)
620	2.516	1886.	16.01	(52.72)
640	2.564	1936.	16.48	(55.42)
660	2.610	1988.	16.94	(58.13)
680	2.655	2041.	17.40	(60.86)
700	2.698	2094.	17.85	(63.60)
720	2.739	2149.	18.30	(66.35)
740	2.779	2204.	18.74	(69.11)
760	2.817	2260.	19.18	(71.89)
780	2.854	2317.	(19.61)	(74.67)
800	2.889	2374.	(20.04)	(77.45)
820	2.923	2432.	(20.46)	(80.24)
840	2.955	2491.	(20.88)	(83.04)
860	2.986	2550.	(21.30)	(85.84)
880	3.016	2610.	(21.70)	(88.64)
900	3.044	2671.	(22.11)	(91.44)
920	3.071	2732.	(22.50)	(94.25)
940	3.097	2794.	(22.89)	(97.06)
960	3.122	2856.	(23.28)	(99.88)
980	3.145	2919.	(23.66)	(102.7)
1000	3.167	2982.	(24.03)	(105.5)

APPENDIX A CONVERSION FACTORS

To convert from	to	multiply by
Pressure in kPa (kN/m^2)	Pressure in Pa	10^3
	Pressure in bar	10^{-2}
	Pressure in atm	$9.869\,233 \times 10^{-3}$
	Pressure in kgf/cm^2	$10.197\,16 \times 10^{-3}$
	Pressure in lbf/in^2	$145.037\,7 \times 10^{-3}$
	Pressure in mmHg	7.500 617
Density in kg/m^3	Density in g/cm^3	10^{-3}
	Density in lb/ft^3	62.428×10^{-3}
Specific enthalpy in kJ/kg	Specific enthalpy in cal/g*	238.846×10^{-3}
	Specific enthalpy in Btu/lb*	429.923×10^{-3}
Specific heat capacity in kJ/kg K	Specific heat capacity in cal/g K*	238.846×10^{-3}
	Specific heat capacity in Btu/lb °F*	238.846×10^{-3}
Dynamic viscosity in μN s/m^2	Dynamic viscosity in N s/m^2	10^{-6}
	Dynamic viscosity in poise or g/cm s	10^{-5}
	Dynamic viscosity in kg/m s	10^{-6}
	Dynamic viscosity in lb/ft h	$2.419\,09 \times 10^{-3}$
	Dynamic viscosity in kgf s/m^2	101.972×10^{-9}
	Dynamic viscosity in lbf s/in^2	145.038×10^{-12}
Dynamic viscosity in mN s/m^2	Dynamic viscosity in N s/m^2	10^{-3}
	Dynamic viscosity in poise or g/cm s	10^{-2}
	Dynamic viscosity in kg/m s	10^{-3}
	Dynamic viscosity in lb/ft h	2.419 09
	Dynamic viscosity in kgf s/m^2	101.972×10^{-6}
	Dynamic viscosity in lbf s/in^2	145.038×10^{-9}
Thermal conductivity in mW/m K	Thermal conductivity in W/m K	10^{-3}
	Thermal conductivity in J/cm s K	10^{-5}
	Thermal conductivity in kcal/m h K*	859.845×10^{-6}
	Thermal conductivity in cal/cm s K*	$2.388\,46 \times 10^{-6}$
	Thermal conductivity in Btu/ft h °F*	577.789×10^{-6}
	Thermal conductivity in Btu in/ft^2 h °F*	$6.933\,47 \times 10^{-3}$
Surface tension in mN/m	Surface tension in N/m	10^{-3}
	Surface tension in J/m^2	10^{-3}
	Surface tension in dyn/cm	1
	Surface tension in erg/cm^2	1
	Surface tension in lbf/ft	68.52×10^{-6}

* Note: the values used for the calorie and British Thermal Unit are International Table values.

APPENDIX B

B1. SPECIFIC HEAT CAPACITIES OF SATURATED LIQUIDS AND VAPOURS

In addition to the constant pressure and constant volume specific heat capacities, there is another specific heat capacity which relates to the saturation line conditions of either phase.

The thermodynamic definitions of the three specific heat capacities are:

$$c_p = \left(\frac{\partial h}{\partial T}\right)_p \ , \tag{B1.1}$$

$$c_v = \left(\frac{\partial u}{\partial T}\right)_v \ , \tag{B1.2}$$

and

$$c_{\text{sat}} = c_p - T\left(\frac{\partial v}{\partial T}\right)_p \left(\frac{\mathrm{d}p}{\mathrm{d}T}\right)_{\text{sat}} \ , \tag{B1.3}$$

where c_p and c_v represent respectively the change in specific enthalpy with temperature at constant pressure and the change in specific internal energy with temperature at constant volume (the values commonly used in engineering design calculations), and

c_{sat} is not $(\partial h/\partial T)_{\text{sat}}$ but is the specific energy required to effect a temperature change while maintaining the fluid in a saturated state, and is the specific heat capacity commonly measured.

The three specific heat capacities are inter-related as follows:

$$c_p = c_v + T\left(\frac{\partial v}{\partial T}\right)_p \left(\frac{\partial p}{\partial T}\right)_v \ , \tag{B1.4}$$

$$\left(\frac{\mathrm{d}h}{\mathrm{d}T}\right)_{\text{sat}} = c_p + \left[v - T\left(\frac{\partial v}{\partial T}\right)_p\right]\left(\frac{\mathrm{d}p}{\mathrm{d}T}\right)_{\text{sat}} \ , \tag{B1.5}$$

$$\left(\frac{\mathrm{d}h}{\mathrm{d}T}\right)_{\text{sat}} = c_{\text{sat}} + v\left(\frac{\mathrm{d}p}{\mathrm{d}T}\right)_{\text{sat}} \ , \tag{B1.6}$$

where $(\mathrm{d}p/\mathrm{d}T)_{\text{sat}}$ represents the change in vapour pressure with temperature.

In practice, the specific heat capacity at saturation conditions cannot be measured by a constant pressure process since a phase change would accompany an energy input and the specific enthalpy of vaporisation and not the specific heat capacity would be determined. For the liquid phase, the specific heat capacities c_p and c_{sat} are in close numerical agreement at low reduced temperatures and differences can usually be neglected. For the saturated vapour, the difference between c_p and c_{sat} should not be ignored.

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89017

Thermophysical properties of propan-2-ol (isopropyl alcohol): Part 1. The ideal gas and the saturated liquid and vapour. ESDU 89017

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ESDU 89017 provides evaluated thermophysical properties data for propan-2-ol (isopropyl alcohol). Equations, based on a critical evaluation of experimental data in the literature are given for the variation with temperature along the saturation line for the following liquid and vapour properties: density, specific enthalpy, specific heat capacity at constant pressure, dynamic viscosity and thermal conductivity. Also given are equations for the variation with temperature along the saturation line of the vapour pressure, latent heat of vaporisation and surface tension. Also presented are equations for the variation with temperature of the specific heat capacity at constant pressure and specific enthalpy in the ideal gas, the dilute gas dynamic viscosity and the dilute gas thermal conductivity.

Saturation line properties in SI units, calculated from the equations are tabulated at 5 K intervals and at 5 degree Celsius intervals from the melting temperature to the critical temperature for the vapour pressure, saturation densities, saturation enthalpies and surface tension and to within at least 56 kelvin of the critical temperature for the remainder. Ideal and dilute gas properties, also in SI units calculated from the equations are tabulated at 20 K and at 20 degree Celsius intervals from the melting temperature to 1000 K. In the absence of experimental data, the tabulated data are extrapolated values and these are indicated in the tables. Conversion factors to facilitate use of the equations and tables in British units are included.

A full bibliography is given of the sources consulted.

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